- ChargeZone Operation Engine (COE) ⚡
  - 🏗️ System Architecture
  - 🧩 Module Directory
    - 🔐 Core Modules
    - 🚗 Session & Session Analysis
    - 🛠️ Infrastructure & Testing
    - 💰 Financials & Support
  - 🚀 Getting Started
    - 📦 Prerequisites
    - 🛠️ Installation & Setup
  - 📑 Available Scripts
    - Backend (/backend)
    - Frontend (/frontend)
  - 🛠️ Technology Stack
  - 🤝 Contributing

ChargeZone Operation Engine (COE) ⚡

Project Status Active | Node.js v18+ | React v19

ChargeZone Operation Engine (COE) is a high-performance operational management suite designed for e-Mobility Service Providers (eMSPs) and Charge Point Operators (CPOs). It provides a unified interface for monitoring charging sessions, managing infrastructure, and handling financial settlements.

🏗️ System Architecture

COE follows a modern decoupled architecture with a focus on domain-driven design.

```
Parse error on line 4:
...>|Mongoose| MongoDB[(MongoDB Atlas/Local
-----------------------^
Expecting 'SPACE', 'GRAPH', 'DIR', 'TAGEND', 'TAGSTART', 'UP',
'DOWN', 'subgraph', 'end', 'MINUS', '--', '==', 'STR', 'STYLE',
'LINKSTYLE', 'CLASSDEF', 'CLASS', 'CLICK', 'DEFAULT', 'NUM',
'PCT', 'COMMA', 'ALPHA', 'COLON', 'BRKT', 'DOT', 'PUNCTUATION',
'UNICODE_TEXT', 'PLUS', 'EQUALS', 'MULT', got 'PS'
```

🧩 Module Directory

The system is organized into specialized modules, each handling a critical aspect of EV charging operations.

🔐 Core Modules

Module	Description	Location
Auth	Secure login with JWT and bcrypt. Role-based access control (Admin vs. Viewer).	`backend/src/modules/auth`
Dashboard	Real-time analytics, KPI tracking, and interactive revenue/session charts.	`frontend/src/pages/Dashboard`
User/Admin	Comprehensive user management, staff profile creation, and permission auditing.	`backend/src/modules/customer-management`

🚚 Session & Session Analysis

Module	Description	Key Features
In-Progress	Live monitoring of currently active charging sessions.	Heartbeat tracking, live meter values.
Faulty Sessions	Automated detection and reporting of failed sessions.	Fault classification, root-cause analysis (RCA).
OCPI Analytics	Deep dive into roaming data and OCPI protocol performance metrics.	Protocol health monitoring, latency tracking.

🛠️ Infrastructure & Testing

Module	Description	Key Features
Charger Simulator	Virtual charger environment for testing backend logic.	OCPP 1.6 support, fault injection, meter simulation.
Station Explorer	Geographic and technical view of all charging stations.	Connectivity monitoring, hardware specs, uptime tracking.
Fleet Management	Tools for B2B partners to manage their corporate EV fleets.	Fleet-specific billing, consumption reports.

💰 Financials & Support

Module	Description	Key Features
CDR Builder	Reconstruct and format Charge Detail Records for accurate billing.	Manual overhaul, data validation, push to CPO.
Settlement	End-of-cycle financial reconciliation for roaming partners.	Clearinghouse integration, margin calculation.
Helpdesk	Internal support system for handling operator and user tickets.	Threaded responses, ticket status (Open/Closed).

🚀 Getting Started

📦 Prerequisites

- **Node.js**: v18.0.0 or higher
- **npm**: v9.0.0 or higher
- **MongoDB**: v6.0 or higher (Local or Atlas)

🛠️ Installation & Setup

1. **Clone the Project**

```
git clone https://github.com/DarshRajput87/ChargeZoneOperationSystem.git
cd ChargeZoneOperationSystem
```

2. **Backend Configuration**

```
cd backend
npm install
cp .env.example .env # Update with your DB URI and JWT Secret
npm run dev
```

3. **Frontend Configuration**

```
cd ../frontend
npm install
npm run dev
```

📑 Available Scripts

Backend (`/backend`)

- `npm run dev`: Start development server with `nodemon`.
- `npm start`: Production server start.
- `npm run setup-indexes`: Initialize MongoDB indexes for optimal performance.

Frontend (`/frontend`)

- `npm run dev`: Launch Vite development server (`localhost:5173`).
- `npm run build`: Production build generation.
- `npm run lint`: Code quality check using ESLint.

🛠️ Technology Stack

- **Frontend**: React 19, Vite, TailwindCSS, Lucide Icons, Recharts, Axios.
- **Backend**: Node.js, Express.js, MongoDB + Mongoose, JWT, Joi, node-cron.
- **Protocols**: OCPI, OCPP 1.6 (via Simulator).

🤝 Contributing

For bug reports or feature requests, please open an issue in the repository or contact the COE maintainers.

Developed by ChargeZone Engineering Team.